

Tutorial 2

- Q1 An aircraft control surface servo-actuator is controlled by a unity feedback control system as indicated in the Figure Q1 below.

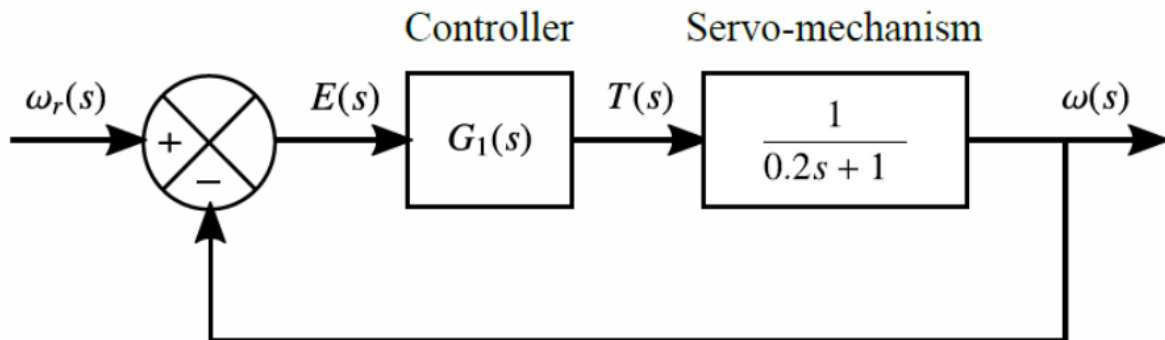


Figure Q1

The angular velocity of the surface is denoted by ω , whilst the reference value is ω_r . The transfer function of the controller is $G_1(s)$.

- If a proportional controller is used what value of gain, K_p is required to ensure a steady state error of 5% due to a step input? [7 marks]
- To eliminate the steady state error an integral controller of gain K_i is used. Determine the conditions on K_i which ensure a non-oscillatory response. [7 marks]
- A PID controller is then applied with the integral gain K_i set to a value of 2. If a closed-loop damping factor of 0.8 is required determine appropriate values for the proportional and derivative gains (K_p and K_d) such that the settling time of the system is 1.5 seconds. [11 marks]

- Q2 A cooling fan's angular speed ω can be related to the applied torque $T_A(t)$ by the relationship:

$$J\dot{\omega} = T_A(t)$$

where J is the moment of inertia of the fan. The applied torque $T_A(t)$ is the difference between the driving torque $T(t)$, supplied by a motor, and a disturbance torque $N(t)$, due to changes in the local airstream. It is desirable to keep the fan running at a constant speed ω_r and so a feedback control system is used to counter the disturbances $N(t)$. The controller has transfer function $G_c(s)$ which operates on the error between the actual fan speed ω and the required speed ω_r . The controller (via an electric servo motor) supplies the driving torque $T(t)$.

- (a) Draw the block diagram of this system. [8 marks]
- (b) If proportional control with gain K_p is used, derive an expression for the steady state error due to a unit step in the disturbance. [8 marks]
- (c) Show that if integral action of gain K_i is added to the proportional action then the steady state error can be removed. [9 marks]